

A GENTLE INTRODUCTION TO MACHINE LEARNING IN R USING TIDYMODELS

Tengku Muhammad Hanis Bin Tengku Mokhtar, PhD
Department of Community Medicine,
Faculty of Medicine,
Universiti Sultan Zainal Abidin (UniSZA)

June 24, 2026

Material



Aims

- Learn an overview of machine learning
- Learn a general step how to do machine learning in R
- The aim is to spark your interest in R and machine learning

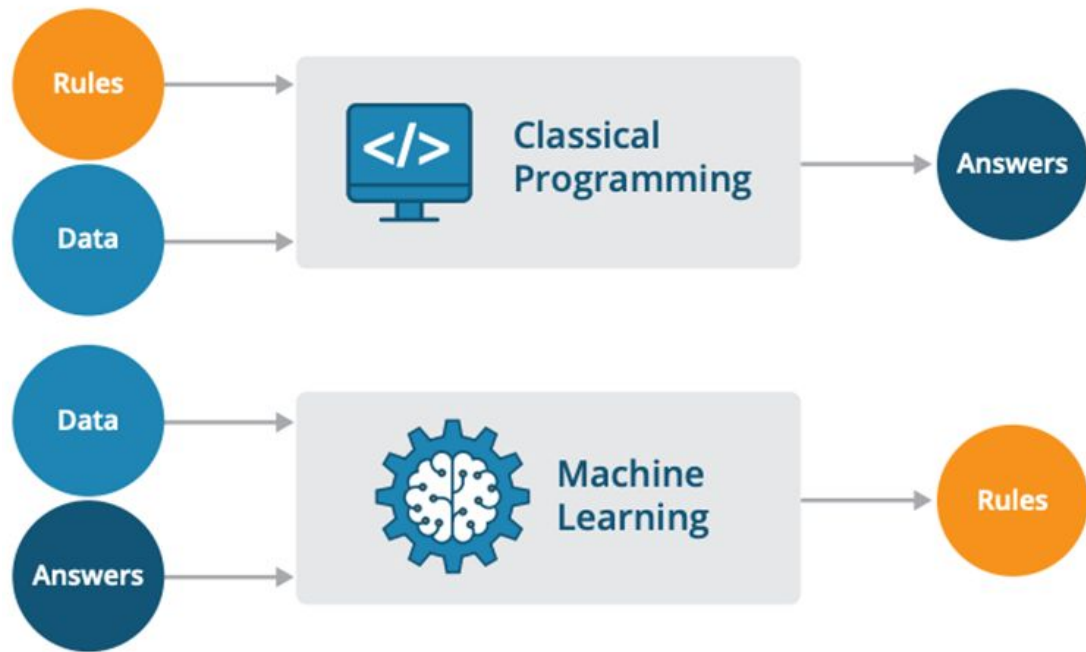
Contents

- [What is ML?](#)
- [ML types](#)
- [Why use R?](#)
- [Intro to R and Posit Cloud](#)
- [What is tidymodels?](#)
- [Step-by-step in tidymodels](#)
- [Hands-on in Posit Cloud](#)
- [What is next?](#)

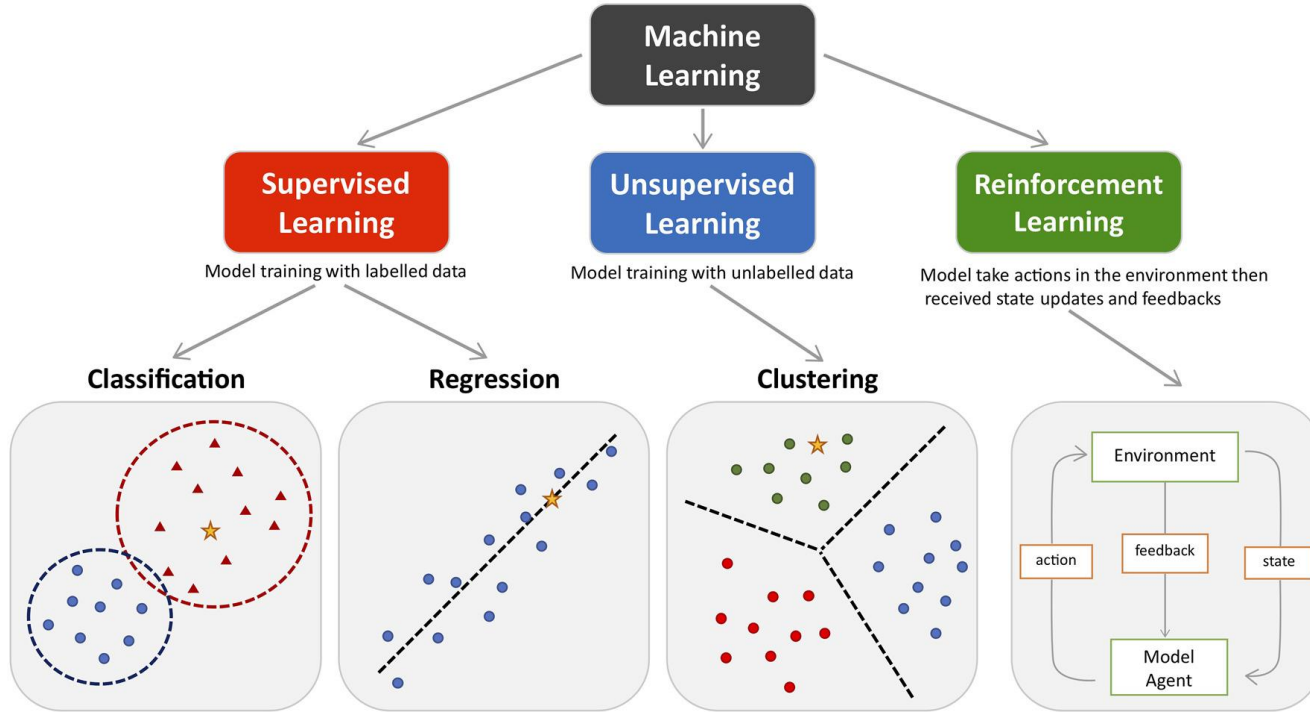
What is ML?

- Machine learning learns pattern from the data to produce the prediction

Traditional programming	ML
You give the computer the exact rules	You give the computer the data and the answers - it figures out the rules for you
Eg: If a patient's blood pressure is over 140, flag them for hypertension	Eg: From the health records of 1,000 patients, and a list of who actually developed hypertension. Find the hidden pattern so we can predict who will get it next.



ML types



Why use R?

- R is a programming language specifically designed for working with data (data analysis)
- Advantages of Learning R:
 - Free and open source
 - Large R community - good support
 - Freely available packages > 20,000 for variety of uses and functions



The actual software



The IDE software



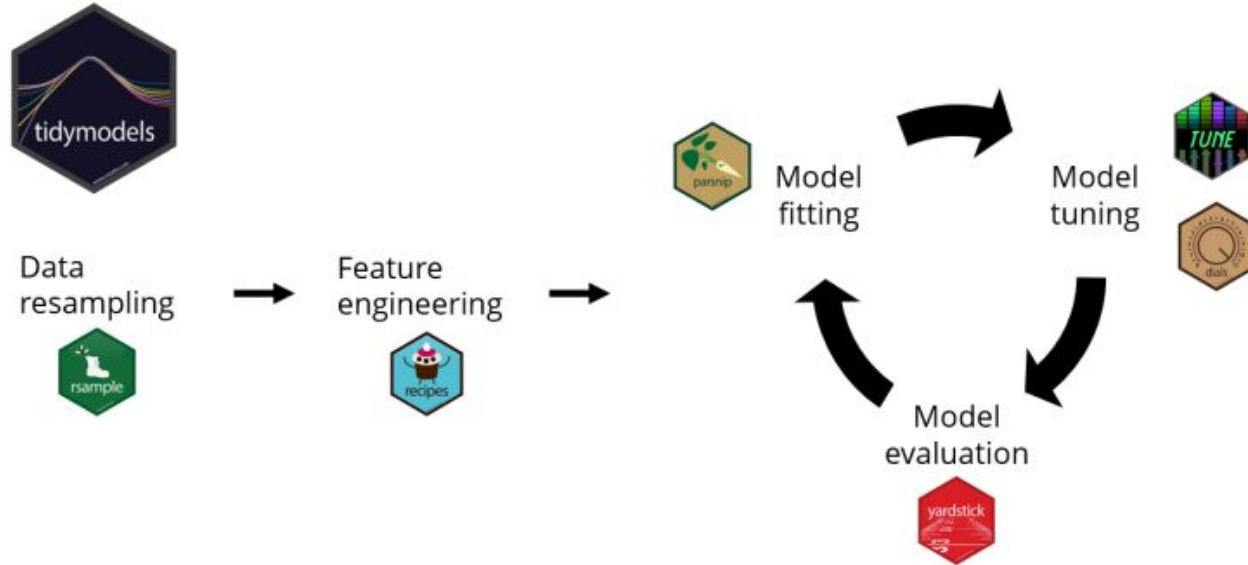
The cloud version

Intro to R and Posit Cloud

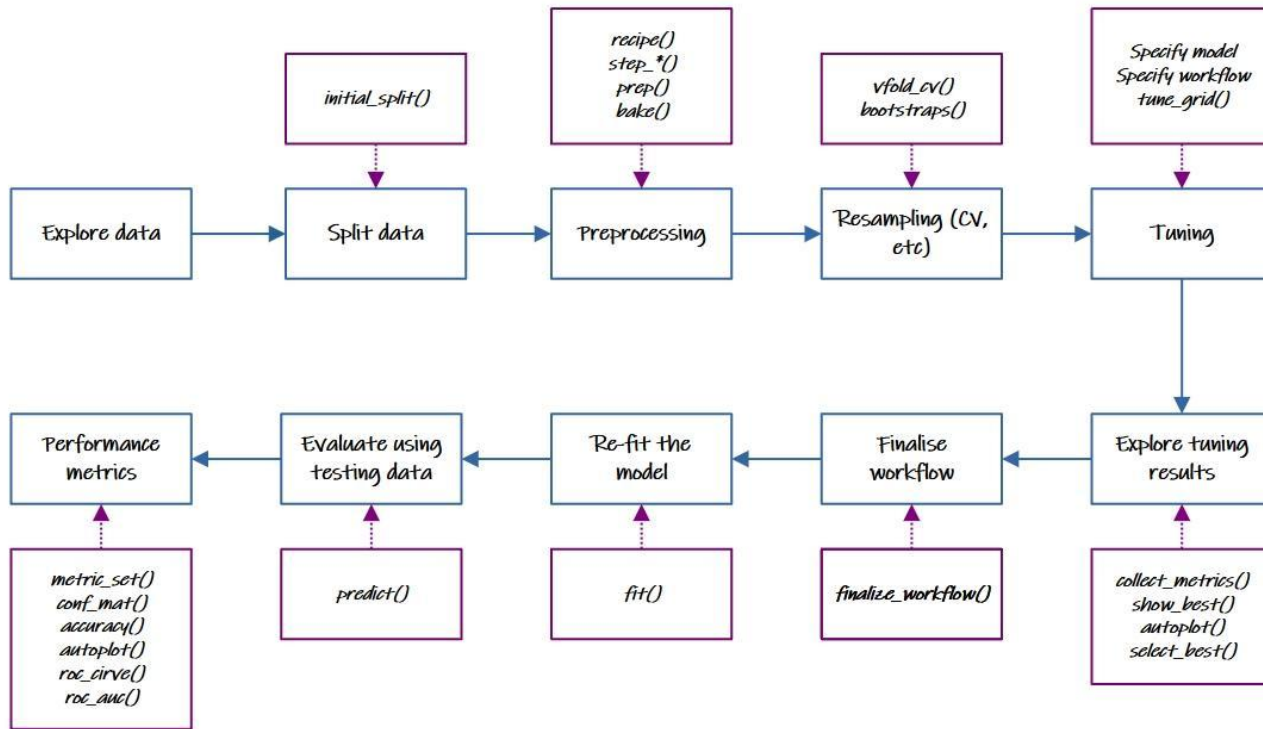
- [Sign up to Posit Cloud](#)
- What is:
 - R script, console
 - Pipe operators: `|>`, `%>%`
 - Packages and installation
 - Get help: ?

What is tidymodels?

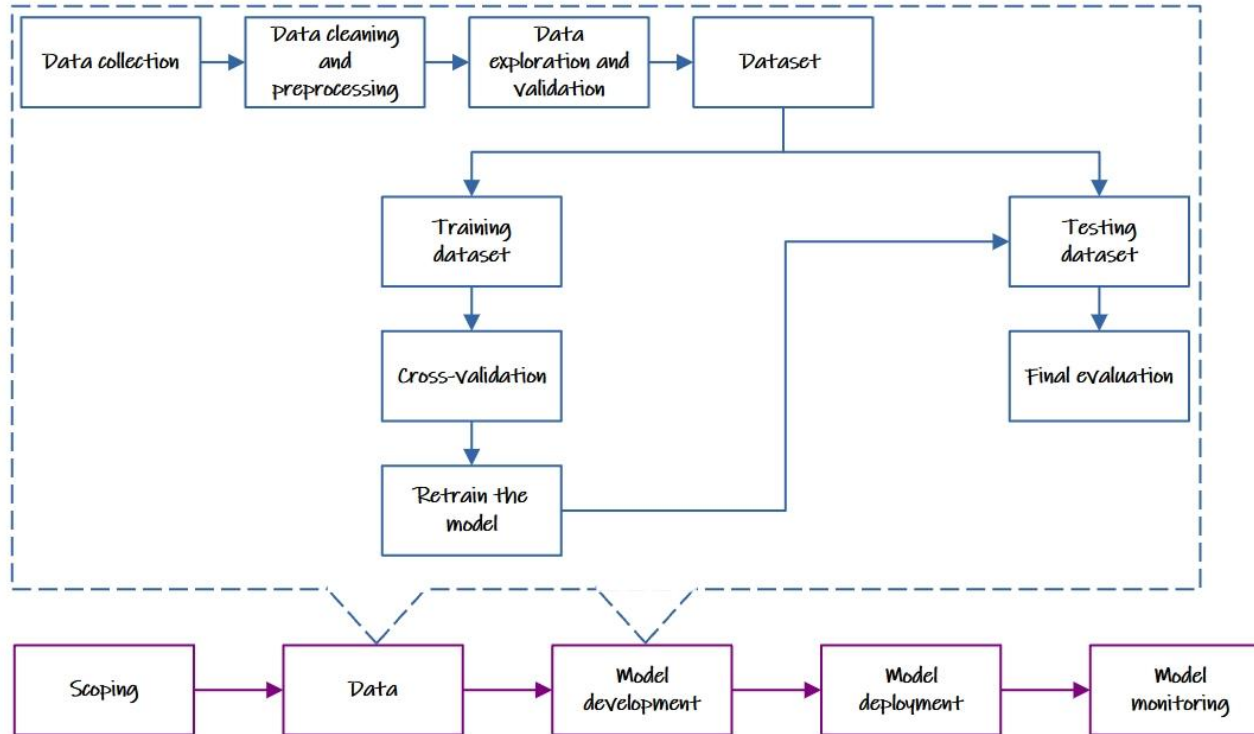
- A collection of R packages for machine learning



Step-by-step in tidymodels



ML workflows (MLOps)

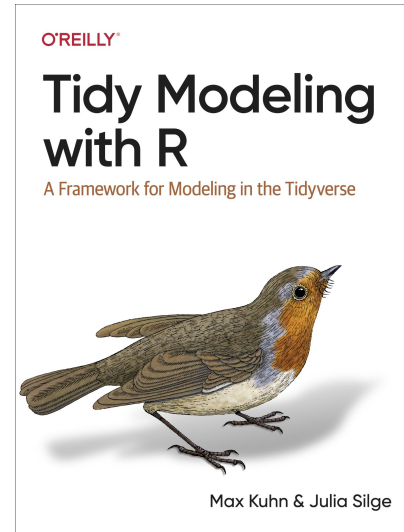
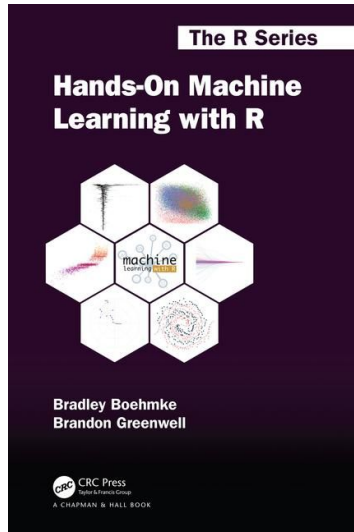
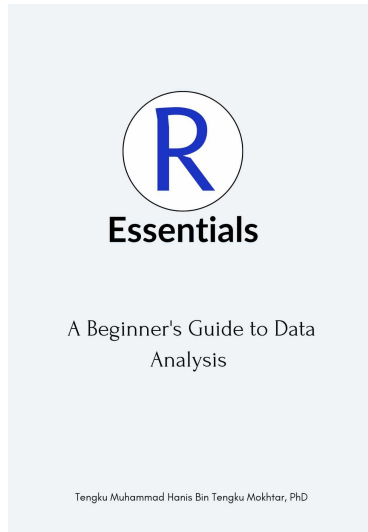


Hands-on in Posit Cloud

- Open Posit Cloud
- Click on “New Project” > “New Project on Git Repository”
- Paste the link: <https://github.com/tengku-hanis/prod2026.git> in “URL of your Git Repository” > Click “OK”

What is next?

- Learn [basics of R](#)
- Learn [machine learning in R](#)
- Learn [tidymodels](#)



Additional resources

- Cheatsheet for [parsnip package](#)
- Cheatsheet for [recipes package](#)
- Many more [cheatsheets](#)



tengkuhanismokhtar@gmail.com
tengkuhanismokhtar@unisza.edu.my
<https://tengkuhanis.netlify.app/>